

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, suspension	Molecules Electroporated	DNA: linearized plasmid DNA; RSV LTR promoting human HLA Class II genes, along with HTK promoter (neo r gene) containing plasmid (co-transfection)
Species Used	Human, K562, chronic myeloid leukemia; U937, histiocytic lymphoma		

Before the Pulse

Cell Growth Medium	RPMI 1640 + 10% Fetal Calf Serum (GIBCO/ BRL, Sigma)	Growth Phase at Harvest	Early log phase growth
		Pre-pulse Incubation	10 min. on ice

Wash Solution RPMI without serum

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature	4 °C	Cuvette Gap	0.4 cm
Electroporation Medium*	RPMI without serum	Voltage	0.2 kV
Cell Density	10 (7) cells / ml	Field Strength	0.5 kV/cm
Volume of Cells	0.4 ml	Capacitor	960 µF
DNA Concentration	1 µg / µl	Resistor	(Pulse Controller) none Ω
DNA Resuspension Buffer	TE (10 mM Tris, 1 mM EDTA, pH 8.0)	Time Constant	15 to 20 msec
Volume of DNA	20 µl		
After the Pulse			
Outgrowth Medium	RPMI 1640 + 10% Fetal Calf Serum		

Relevant Publications and/or Comments

Note: exponential values designated in parentheses. Cells resistant to G418 (1µg/ml) grew out of post-pulse cell culture after about 15-30 days, with low % of survival. G418 resistant cells were then dilution-cloned in preparation for screening a Class II expressing clone. However, before the majority of cells could be screened, the clones were lost. I don't know how efficient electroporation procedure was since we were trying to get expression of a two-chain molecule on the cell surface - many factors could go wrong post-electroporation but prior to protein expression on the cell surface.

Outgrowth Temperature	37 °C
Length of Incubation	15 to 30 days
Selection Method or Assay Used	G418 resistance
Electroporation Efficiency	Unknown (see notes)
Per Cent Survival	<1%

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Survey Number

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